Индивидуальный проект №2

Основы информационной безопасности

Кузьмина Мария Константиновна

Цель работы

Установка DVWA

Установка

Обновляем список пакетов, устанавливаем необходимые компоненты

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.4
MB]
68% [3 Contents-amd64 29.4 MB/53.4 MB 55%]
73% [3 Contents-amd64 34.2 MB/53.4 MB 64%]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [274 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [185 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [880 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [14.3 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [33.8 kB]
Fetched 76.0 MB in 10s (4,252 kB/s)
1788 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(kali㉿kali)-[~]
└─$ sudo apt install -y apache2 mariadb-server mariadb-client php php-mysqli
php-gd php-xml php-mbstring git unzip wget
Note, selecting 'php8.4-mysql' instead of 'php-mysqli'
unzip is already the newest version (6.0-29).
unzip set to manually installed.
wget is already the newest version (1.25.0-2).
The following packages were automatically installed and are no longer require
d:
  libconfig-inifiles-perl  libcrypt-dev
Use 'sudo apt autoremove' to remove them.

Upgrading:
  apache2                  mariadb-common
  apache2-bin              mariadb-plugin-provider-bzip2
  apache2-data             mariadb-plugin-provider-lz4
  apache2-utils            mariadb-plugin-provider-lzma
  git                      mariadb-plugin-provider-lzo
  git-man                  mariadb-plugin-provider-snappy
  libapache2-mod-php8.4    mariadb-server
  libc-bin                 mariadb-server-compat
  libc-dev-bin             mariadb-server-core
  libc-l10n                php
  libc6                    php8.4-cli
  libc6-dev                php8.4-common
  libc6-i386               php8.4-mysql
  locales                  php8.4-opcache
  mariadb-client           php8.4-readline
  mariadb-client-core

Installing:
  php-gd  php-mbstring  php-xml

Installing dependencies:
  libc-gconv-modules-extra  php8.4-gd  php8.4-mbstring  php8.4-xml
```

Запуск сервисов Apache и MySQL

Запускаем Apache, запускаем Mariadb(MySQL), проверяем, что оба сервиса работают

```
kali@kali: /var/www/html/DVWA/config
Session  Actions  Edit  View  Help
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for base-files (1:2025.4.0) ...
Processing triggers for php8.4-cli (8.4.16-1+b2) ...
Processing triggers for libapache2-mod-php8.4 (8.4.16-1+b2) ...

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start apache2

┌──(kali㉿kali)-[~]
└─$ sudo systemctl enable  apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/lib/systemd/system/apache2.service'.

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start mariadb

┌──(kali㉿kali)-[~]
└─$ sudo systemctl enable  mariadb
Synchronizing state of mariadb.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mariadb
Created symlink '/etc/systemd/system/multi-user.target.wants/mariadb.service' → '/usr/lib/systemd/system/mariadb.service'.

┌──(kali㉿kali)-[~]
└─$ sudo systemctl status apache2 --no- pager
systemctl: option '--no-' is ambiguous; possibilities: '--no-block' '--no-legend' '--no-pager' '--no-wall' '--no-warn' '--no-reload' '--no-ask-password'

┌──(kali㉿kali)-[~]
└─$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-03-21 08:12:56 EDT; 1min 49s ago
 Invocation: cddcb7a07961a65b4a9b9637615664
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 17552 (apache2)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 6 (limit: 2116)
     Memory: 26.5M (peak: 28.7M)
        CPU: 110ms
     CGroup: /system.slice/apache2.service
             ├─17552 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─17556 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─17557 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─17559 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─17560 /usr/sbin/apache2 -k start -DFOREGROUND
             └─17561 /usr/sbin/apache2 -k start -DFOREGROUND

Mar 21 08:12:56 kali systemd[1]: Starting apache2.service - The Apache HTTP Server ...
Mar 21 08:12:56 kali apachectl[17552]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Mar 21 08:12:56 kali systemd[1]: Started apache2.service - The Apache HTTP Server.

┌──(kali㉿kali)-[~]
└─$ sudo systemctl status mariadb --no-pager
● mariadb.service - MariaDB 11.8.6 database server
```

Клонирование репозитория, настройка прав доступа, конфигурационного файла

Переходим в директорию /var/www/html/, клонируем репозиторий, устанавливаем правильные права. В редакторе находим строки, отвечающие за подключение к базе данных данные, изменяем данные на те, которые указывали ранее (имя, пароль)

```
kali@kali: /var/www/html/DVWA/config
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[/var/www/html]
└─$ sudo rm -rf index.html

┌──(kali㉿kali)-[/var/www/html]
└─$ sudo git clone https://github.com/digininja/DVWA.git
Cloning into 'DVWA' ...
remote: Enumerating objects: 5703, done.
remote: Counting objects: 100% (52/52), done.
remote: Compressing objects: 100% (35/35), done.
remote: Total 5703 (delta 27), reused 25 (delta 17), pack-reused 5651 (from 2)
Receiving objects: 100% (5703/5703), 2.74 MiB | 796.00 KiB/s, done.
Resolving deltas: 100% (2836/2836), done.

┌──(kali㉿kali)-[/var/www/html]
└─$ cd /var/www/html/DVWA

┌──(kali㉿kali)-[/var/www/html/DVWA]
└─$ sudo chown -R www-data:www-data /var/www/html/DVWA

┌──(kali㉿kali)-[/var/www/html/DVWA]
└─$ sudo chmod -R 755 /var/www/html/DVWA

┌──(kali㉿kali)-[/var/www/html/DVWA]
└─$ sudo chmod -R 777 hackable/uploads/

┌──(kali㉿kali)-[/var/www/html/DVWA]
└─$ sudo chmod 777 config/

┌──(kali㉿kali)-[/var/www/html/DVWA]
└─$ cd /var/www/html/DVWA/config

┌──(kali㉿kali)-[/var/www/html/DVWA/config]
└─$ sudo cp config.inc.php.dist config.inc.php

┌──(kali㉿kali)-[/var/www/html/DVWA/config]
└─$ sudo nano config.inc.php

┌──(kali㉿kali)-[/var/www/html/DVWA/config]
└─$ sudo nano /etc/php/8.2/apache2/php.ini

┌──(kali㉿kali)-[/var/www/html/DVWA/config]
└─$ php -v
PHP 8.4.16 (cli) (built: Mar  2 2026 16:15:02) (NTS)
```

Инициализация DVWA через веб-интерфейс

Открываем браузер, переходим по ссылке http://127.0.0.1/DVWA/, входим с учетными данными по умолчанию, устанавливаем уровень безопасности (пока low), переходим в раздел ‘SQL Injection’, в поле вводим ‘or 1=1’, появляются ссылки на веб-страницы

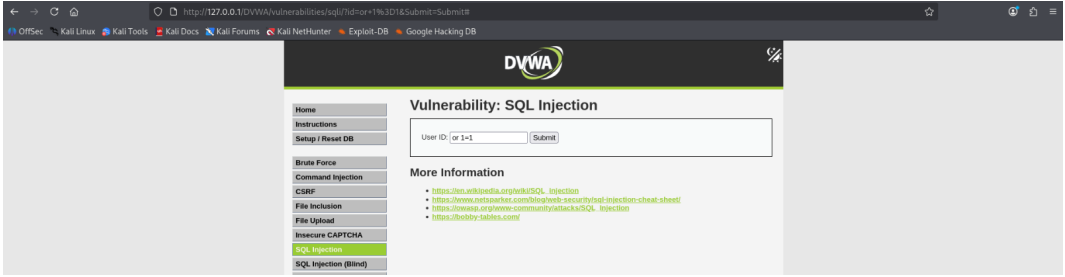

Figure 4: снимок экрана

Выводы

Установка прошла успешно